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MISSOURI UNIVERSITY OF SCIENCE AND TECHNOLOGY
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Section 5.3

Diagonalization

Similarity of Matrices

If A and B are $n \times n$, then A is similar to B if there is an invertible matrix P such that $A = PBP^{-1}$.

If A and B are similar, then they have the same characteristic polynomial (and, thus, the same eigenvalues with the same multiplicities).

Cautions about Similarity

Two matrices with the same eigenvalues may or may not be similar.

Example:

$\begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$ and $\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$ are not similar.

Similarity is not the same as row equivalence. Row operations on a matrix usually change its eigenvalues.

Diagonalization

A square matrix A is said to be diagonalizable if it is similar to some diagonal matrix D ; that is, if $A = PDP^{-1}$ for some invertible matrix P .

Example 1

Let $A = PDP^{-1}$ and use this to compute A^4 .

$$P = \begin{bmatrix} 2 & -3 \\ -3 & 5 \end{bmatrix}$$

$$D = \begin{bmatrix} 1 & 0 \\ 0 & 1/2 \end{bmatrix}$$

The Diagonalization Theorem

An $n \times n$ matrix A is diagonalizable if and only if A has n linearly independent eigenvectors.

$A = PDP^{-1}$ with D diagonal if and only if the columns of P are linearly independent eigenvectors of A . In this case, the diagonal entries of D are the eigenvalues of A that correspond to the eigenvectors in P .

The Diagonalization Theorem

Equivalently, we can say an $n \times n$ matrix A is diagonalizable if and only if there are enough eigenvectors to form a basis for $\mathbb{R}^n$. Such a basis is called an eigenvector basis of $\mathbb{R}^n$.

Example 2

Diagonalize each matrix, if possible.

$$A = \begin{bmatrix} 5 & 1 \\ 0 & 5 \end{bmatrix}$$

$$B = \begin{bmatrix} 2 & 3 \\ 4 & 1 \end{bmatrix}$$

Example 3

Diagonalize each matrix, if possible.

$$A = \begin{bmatrix} 2 & 2 & -1 \\ 1 & 3 & -1 \\ -1 & -2 & 2 \end{bmatrix}; \lambda = 1, 5$$

$$B = \begin{bmatrix} 4 & 0 & 0 \\ 1 & 4 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

Theorem

An $n \times n$ matrix with n distinct eigenvalues is diagonalizable.

Note that this is a *sufficient* but not *necessary* condition.

Example 4

Determine if each statement is true or false. Assume all matrices are $n \times n$.

- a) A is diagonalizable if $A = PDP^{-1}$ for some matrix D and some invertible matrix P .
- b) If $\mathbb{R}^n$ has a basis of eigenvectors of A , then A is diagonalizable.
- c) A is diagonalizable if and only if A has n eigenvalues, counting multiplicities.
- d) If A is diagonalizable, then A is invertible.

Theorem

An $n \times n$ matrix A is diagonalizable if and only if the sum of the dimensions of the eigenspaces equals n .

This happens if and only if

- i. The characteristic polynomial factors completely into linear factors; and
- ii. The dimension of the eigenspace for each eigenvalue equals the multiplicity of the eigenvalue.

Example 5

Let A be a 4×4 matrix with three eigenvalues. One eigenspace is one-dimensional, and one of the other eigenspaces is two-dimensional. Is it possible that A is not diagonalizable?
